UNIVERSITY OF OTAGO EXAMINATIONS 2019

COMPUTER SCIENCE

COSC343

ARTIFICIAL INTELLIGENCE

Semester 1

(TIME ALLOWED: THREE HOURS)

This examination comprises 7 pages.

Candidates should answer questions as follows:

Candidates should answer **All** questions.

Questions are worth 12 marks and sub marks are shown thus: (2)

The total number of marks for this exam is 60.

The following material is provided:

Nil.

Use of calculators

Any model of calculator provided this is battery powered, silent, truly portable and free of communication capabilities.

Candidates are permitted copies of:

Nil.

TURN OVER

1. Learning

(a) What's the difference between an *episodic* and *sequential* environment? Give an example of each. (2)

(b) What's the difference between *supervised*, *unsupervised* and *reinforcement* learning? Give an example of a task best suited for each. (3)

(c) What do the evaluations of the cost function on the *training* vs. the *test* datasets have to do with *consistency* and *generalisation* of the model's hypothesis? (2)

(d) Given an update rule $\mathbf{w} := \mathbf{w} + \alpha\Delta\mathbf{w}$ for the *steepest gradients descent* optimisation, where $\mathbf{w}$ is a vector of model parameters and $\Delta\mathbf{w}$ is the negative gradient of some cost function with respect to parameters, what is the role of the α in this equation? What is the general guideline for setting the value of α and how does this choice affect the dynamics of the optimisation? (3)

(e) Does the *steepest gradient descent* optimisation always find the global minimum of the cost function? Are there any conditions under which this behaviour with respect to guarantees of finding the global minima would change? (2)

2. Neural networks

Definitions of the hard-limiting $\sigma_h(v)$, sigmoid $\sigma_s(v)$ and relu $\sigma_r(v)$ activation functions:

$$\sigma_h(v) = \begin{cases} 1 & v > 0 \\ 0 & v \leq 0 \end{cases} \qquad \sigma_s(v) = \frac{1}{1+e^{-v}} \qquad \sigma_r(v) = \begin{cases} v & v > 0 \\ 0 & v \leq 0 \end{cases}$$

(a) What is the difference between the *perceptron learning rule* and *backpropagation*? (2)

(b) What is the geometric interpretation for the role of the weights and biases of an artificial neuron with a hard-limit function in the input space? (3)

(c) Given a set of $16 \times 16 \times 3$ colour images with corresponding labels indicating that there is either a car, a truck or a plane in the image, design a *multilayer perceptron neural network* that (at least in theory) should be capable (once trained) of identifying the objects in the images. Specify the number of inputs, layers, neurons per layer, choices of activation functions, and cost function suitable for this problem. Indicate which aspects of your design are dictated by the data, and which ones are arbitrary choices. Although for the arbitrary choices there is no single right answer, still make your choices and justify them. (3)

(d) Is a *recurrent neural network* better suited for the task described in 2c? Why or why not? (2)

(e) Is a *convolutional neural network* better suited for the task described in 2c? Why or why not? (2)

3. Uncertainty

You're given the following set of subject lines from e-mail messages, each labelled by the recipient as either spam or ham:

"garage rolex sale"	spam
"sale notice"	spam
"watch the final"	ham
"final notice"	ham
"watch sale"	spam
"garage sale on sunday"	ham
"rolex watch"	spam

(a) What is the probability of the word "garage" in any subject line? (1)

(b) In the table above, what is the probability of the word "garage" in a spam message? (1)

(c) From the definition of conditional probability we have $P(A, B) = P(A|B)P(B) = P(B|A)P(A)$. Derive Bayes' rule from this and explain how it is useful when trying to compute the probability of a cause given effects. (3)

(d) When applying a Naive Bayes Classifier to spam filtering, what assumption is being made about the words in the messages? Is this assumption correct? (2)

(e) Based on the training data above, show the Naive Bayes Classifier formula and computation that determines the label for a new message with the subject title "garage notice". Show the formula you used and the calculation. (3)

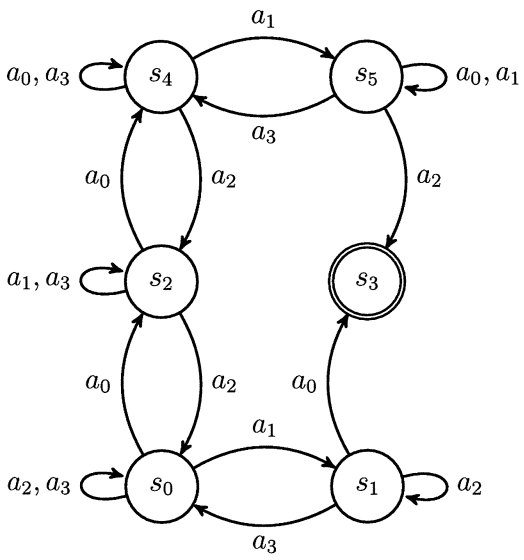
(f) What can *entropy* tell us about a split of labelled data into distinct groups and how is this information used in decisions trees? (2)

4. Reinforcement learning

Consider an environment of 6 discrete states $s_0, \ldots, s_5$, four actions $a_0, \ldots, a_3$, and the reward function:

$$R(s) = \begin{cases} 1 & \text{when } s = s_3 \\ -0.04 & \text{otherwise.} \end{cases}$$

The environment is governed by the deterministic transition model and the agent by a policy given respectively in the following state machine and Q-table:



State machine

	a_0	a_1	a_2	a_3
s_0	-0.01	0.90	0.03	0.02
s_1	0.99	0.04	0.02	0.01
s_2	-0.02	0.02	0.43	-0.00
s_3	1.00	1.00	1.00	1.00
s_4	-0.01	0.00	0.01	-0.01
s_5	0.00	0.09	0.15	0.02

Q-table

(a) What is an *optimal policy*? (1)

(b) Assuming initial state s_0 and terminal state s_3, what is the sequence of actions of the agent according to the policy given in the Q-table above? Assuming no discount on the future reward, what's the reward for this sequence of actions? Is this policy optimal for this starting state? If not, what changes to the table would need to be made to make it optimal and what would the reward be then? (4)

(c) Assuming initial state s_4 and terminal state s_3, what is the sequence of actions of the agent according to the policy given in the Q-table above? Assuming no discount on the future reward, what's the reward for this sequence of actions? Is this policy optimal for this starting state? If not, what changes to the table would need to be made to make it optimal and what would the reward be then? (4)

(d) If the environment was different such that

$$R(s) = \begin{cases} 1 & \text{when } s = s_3 \\ 0 & \text{otherwise,} \end{cases}$$

what could be done during training in order to encourage the agent to traverse the least number of states from its initial state to the terminal one? (1)

(e) What is the general technique in reinforcement learning that gives the agent a chance to discover the best solution given the existence of several different ways of solving a problem? Explain how this technique works. (2)

5. Adversarial search

Consider the following Tic-tac-toe board states and an agent playing as 'O':

X		
	O	
		X

State 1

X	O	
	O	
		X

State 2

(a) Calculate the agent's evaluation of the two above states according to function

$$e = \begin{cases} \infty & \text{, win for MAX} \\ -\infty & \text{, win for MIN} \\ l_{\text{MAX}} - l_{\text{MIN}} & \text{, otherwise} \end{cases},$$

where l_{MAX} is the number of possible winning lines for MAX on the board, and l_{MIN} is the number of possible winning lines for MIN on the board. (2)

(b) Show the depth limited *minimax* search tree for the agent considering its next move from State 1. Limit the search depth to 2 and use the evaluation function given above to determine the values at the leaf states. Including only rotationally invariant states in your search tree is fine. Clearly mark each ply of the search tree as MIN or MAX, and show the values of each state. Show the state of the board after the move chosen by the agent. (8)

(c) Is the move chosen by the agent in your answer from 5b a good move? If not, what is the reason that minimax does not give an optimal move? If you are unable to complete question 5b, take State 2 shown above as the result of agent's choice and answer this question. (2)

END

COSC 343

PLEASE DO NOT

TURN OVER YOUR

EXAMINATION

PAPER UNTIL

INSTRUCTED TO

DO SO